A Comparison of ML Methods for AMR Gene Class Prediction in FMT Patients

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Abstract

Antimicrobial resistance (AMR), a major modern healthcare problem, can be limited by supplementing heavy antibiotic regimes with treatments called fecal microbiota transplants (FMTs) in the case of patients for whom antibiotics have become ineffective. Machine learning (ML) methods have emerged as a potentially fast and effective way to visualize and predict AMR gene dynamics, though there have been few direct comparisons between different machine learning methods within this application. This research aims to establish direct comparisons between different machine learning methods in analyzing or predicting frequencies of antimicrobial resistance genes in post-FMT patients. To do this, we trained various supervised ML methods (Random Forest, XGBoost, and Linear Regression/Lasso) to predict post-FMT AMR gene frequencies from preprocessed FMT patient datasets. Additionally, unsupervised ML models like principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE) were used for exploratory analysis and visualization of gene clusters. Using the hyperparameter optimization framework Optuna, we determined the optimal hyperparameters for each ML model and evaluated the most accurate supervised ML method for each gene class. For both supervised and unsupervised models, SHapley Additive exPlanations (SHAP) was used to determine feature importance and model interpretability. Preliminary results suggest that Random Forest, while commonly regarded as a strong baseline in machine learning, performed less accurately than other models in predicting AMR gene abundance. This research aims to provide groundwork for further comparison of ML methods in an AMR space and background for the usage of the most optimal ML methods in predicting AMR gene frequencies.

Background

Prolific overuse of antimicrobial medication has spurred the development of antimicrobial resistance genes in microbes, rendering antimicrobial treatments inert. This issue of antimicrobial resistance (AMR) has been widely declared a major global health problem. Bacterial AMR was attributed to an estimated 4.71 million deaths in 2021 alone– a number that could nearly double by 2050. (Naghavi et al.) In order to circumvent patients hosting resistant microbiota and/or prevent continued proliferation of AMR, physicians must seek alternative methods of treatment to traditional antibiotics and antivirals.

Fecal microbiota transplants (FMT) have been found to be one such treatment that reduces AMR mechanisms– particularly the development of multidrug-resistant organisms (Woodworth et al.). FMT treatments exploit bacterial strain tendencies to compete with each other which overall reduce populations of resistant bacteria. This is an incredibly promising method to slow the development of AMR.

However, the development and expression of AMR genes is still steeped in undocumented complexity. Machine learning techniques have emerged as a potential tool for quantifying and predicting AMR at a genomic level. Machine learning (ML) refers to the field of artificial intelligence focusing on the development of algorithms that can learn to solve predictive problems without being explicitly programmed to. Models are typically parameterized on training data and applied to testing data where performance can be evaluated.

While a plethora of studies (Niño et al.; Nguyen et al.; Wang et al.) have shown different ML methods as fast and effective ways to visualize and predict AMR gene dynamics; there are no established best practices and few direct comparisons between different machine learning methods within this application. Even more so is the lack of prior studies involving regression problems, as the most documentation of ML methods and data preprocessing has been for classification (Branco et al.). To amend this knowledge gap, we built and optimized several different regression-based ML models that can predict frequencies of AMR genes in FMT patients. This study aims to establish direct comparisons between these different ML methods, as well as provide groundwork for future evaluations of ML methods in an AMR space.

Methods

Machine Learning Method Selection

To determine the Machine Learning methods tested, we reviewed literature containing summaries of regression machine learning applications in gut microbiome research and focused specifically with those used commonly in phenotypic prediction regression problems. Supervised methods chosen to be compared include Random Forest, XGBoost, and Least Absolute Shrinkage and Selection Operator (LASSO). Unsupervised methods chosen to be compared include Principal Component Analysis (PCA) and T-distributed Stochastic Neighbor Embedding (t-SNE). Detailed explanation and justification of methods selected can be found in Appendix 1.

Data Cleaning & Preprocessing

The dataset was compiled and harmonized from multiple FMT studies involving antimicrobial resistance, as described in Ianiro et al. (2022). We chose to use FMT studies because they provide a dynamic data set for our machine learning models to make predictions on as patient data is taken at various time points throughout the treatment. Two interlinked datasets were retrieved– one containing FMT patient metadata (referred to as "predictor data") and the other containing AMR gene frequency data for each patient sample (referred to as "counts data"). Categorical predictor data was manually converted to representative integers as described in Appendix 2 and in the predictor data spreadsheet notes (Predictors.xlsx). FMT donor predictor data was removed, as well as redundant predictor columns. Each timepoint for a patient was counted as a separate sample. AMR frequency counts data was consolidated by gene class for each sample.

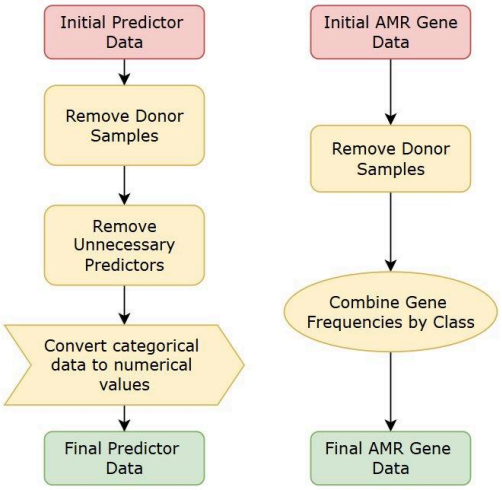

Figure 1: Flow chart presenting the sequence of data cleaning steps.

For the unsupervised methods of PCA and t-SNE, the AMR counts data was additionally modified. All zero data was replaced by 0.000001 to allow normalization using the center log ratio (CLR) transformation function clr() from the scikit-bio package. The counts data was next transformed into euclidean distances using the euclidean_distances() function from the sklearn package. These changes were done to increase the quality of the counts dataset and optimize the potential of these machine learning techniques (Bala and Behal).

Model Comparison, Hyperparameter Optimization & Feature Importance

We used Optuna, an automatic hyperparameter optimization software framework developed for machine learning to determine the optimal model hyperparameters and compare models within each AMR gene class. The hyperparameters selected to be varied depended on the type of model, and can be seen in the code linked in *Software Availability.* We selected the coefficient of determination (r-squared value) to evaluate individual mode performance, as it is more interpretable than other metrics for regression problems (Chicco et al.). Individual feature performance was modeled by SHapley Additive exPlanations (SHAP), a game theoretic approach to explaining ML model outputs. Feature importance across gene classes was also modeled using a SHAP beeswarm plot.

Supervised Model Development

All models were created in Python's Jupyter Notebooks using the Python 3 ipykernel on Purdue University's Gautschi servers.

- Iteration: The following steps were repeated for every study using every gene class for each supervised machine learning model in the counts data.
- Optimization: Model parameters were suggested by an Optuna study for each trial using Optuna's trial.suggest_int or suggest_float for each relevant hyperparameter. 50 trials were performed for each study.
- Data Division: Using the Optuna suggested proportion, data was split into testing and training data using sklearn's train_test_split() function.
- Modeling: Supervised models were initialized using the sklearn (RandomForestRegressor and linear_model.Lasso functions for Random Forest and Lasso respectively) and XGBoost packages (XGBRegressor function for XGBoost). Models were then fit to the training data.
- Comparison: The coefficient correlation (r-squared) values were calculated by using the models to predict the testing data.
- Replication: The models with the highest r-squared values and their respective hyperparameters were then saved to .joblib files. The random seed was set to 0 for all models for reproducibility.

Unsupervised Model Development

All models were created in Python's Jupyter Notebooks using the Python 3 ipykernel on Purdue University's Gautschi servers.

- Modeling: Using the PCA and TSNE functions within the sklearn package (for PCA and t-SNE models respectively), models were trained on the modified counts data (See *Data Cleaning & Preprocessing*)
    - PCA Parameters: n_components = 2
    - t-SNE Parameters: n_components=2, verbose=1, max_iter=5000. Five t-SNE models were created with varying perplexity (5, 15, 25, 35, and 45).
- Visualization: Models were plotted and saved using the seaborne and matplotlib packages.
    - t-SNE: Color coded plots were created for each feature.

Software Availability

The following Python software libraries were utilized:

- Pandas: Used for importing data and general data structure manipulation.
- Numpy: Used for data structure manipulation.
- Scikit-learn: Used for implementing various ML algorithms (RandomForestRegressor, linear_regression.Lasso, PCA, and TSNE) and data preprocessing (euclidean_distances).
- Scikit-bio: Used for center log ratio data preprocessing (clr).
- XGBoost: Used for implementing XGBoost model algorithms (XGBRegressor).
- Optuna: Used for hyperparameter optimization & model comparison.
- SHAP: Used for feature importance modeling and data visualization.
- Matplotlib & Seaborn: Used for data visualization of plots and graphs.
- Joblib: Used to save and load models and trials for repetition.

To ensure the reproducibility of the study, all code is documented in the Slizovskiy Lab github: https://github.com/SlizovskiyLab/RebekahNotebook

Dataset Accession

Data can be made available upon reasonable request.

Results

Supervised Model Comparison

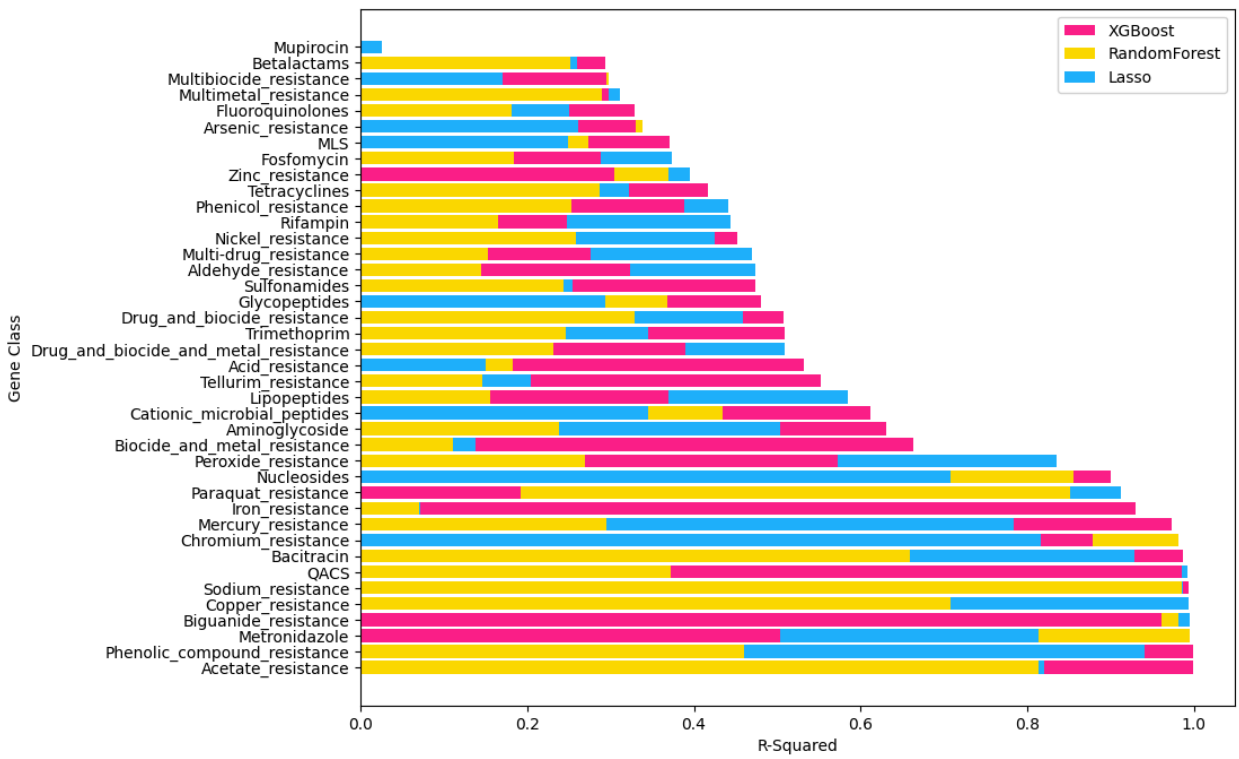

Figure 2: Overall Comparison of Supervised Machine Learning Model Correlation Coefficients for FMT Patient Gene Classes. Models are compared via the absolute values of their r-squared values displayed on the x-axis and sorted from least to greatest r-squared overall. The Y-axis labels the gene classes that each model is predicting the frequency of.

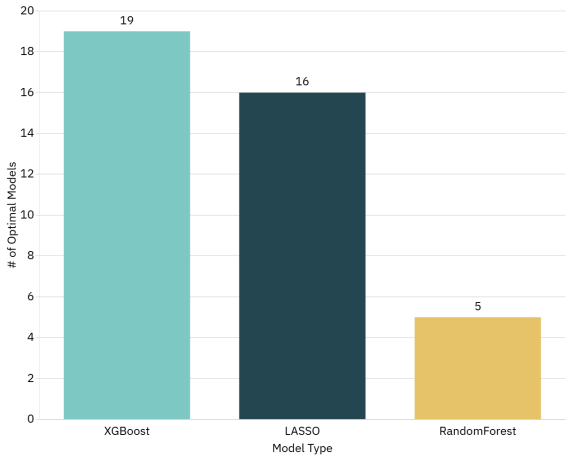

Figure 3: ML Model Type Comparison Graph. Coefficient Correlations were tallied by model type and depicted in a bar graph for easy visual identification.

Of the models, the most gene classes were best represented by XGBooster, then LASSO, and finally, Random Forest. Overall, the mupirocin gene class was the most difficult to represent using supervised machine learning models, as shown by its consistently low correlation coefficient values (For XGBoost, r^2 = 0.02500009536743164, for RandomForest, r^2 = 0.025000000000000133, and for Lasso, r^2 = 0.025000000000000133). On the other hand, acetate resistance displayed the highest and most consistent correlation coefficient values across all three supervised machine learning models.

Random Forest Models

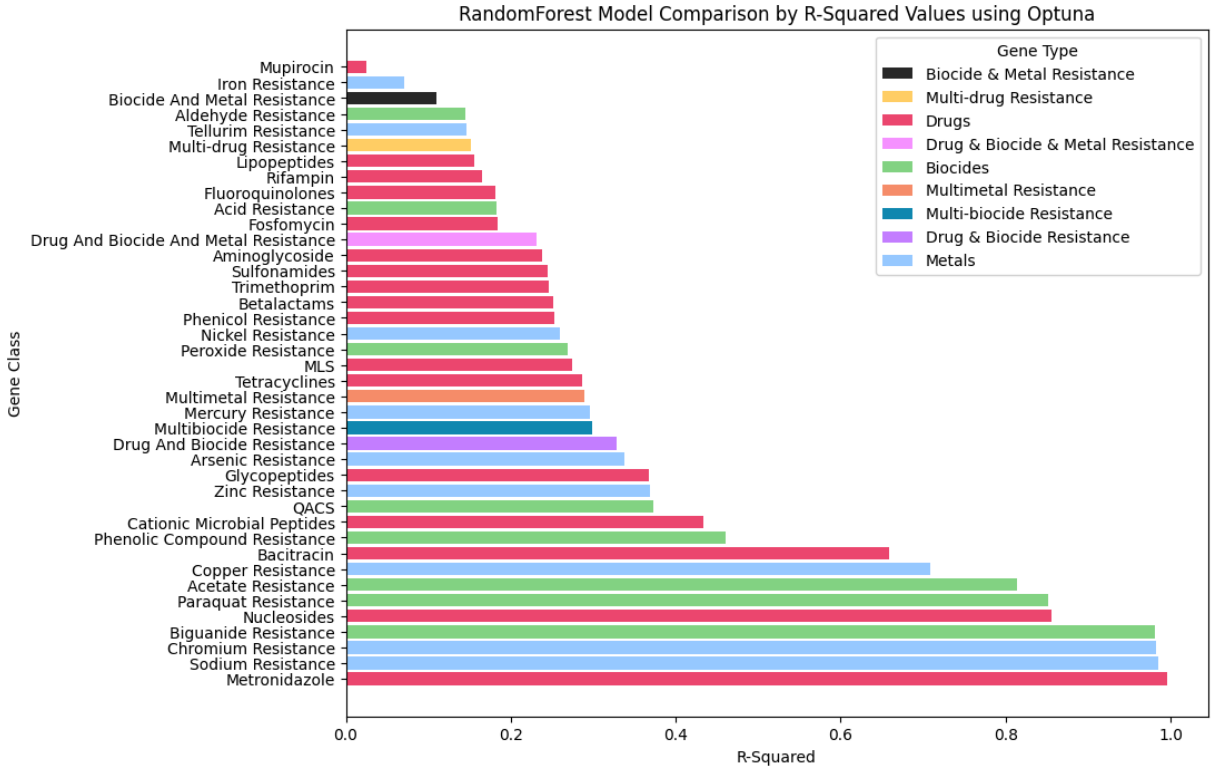

Figure 4: R-squared Values for Random Forest Models. Models are compared via the absolute values of their r-squared values displayed on the x-axis and sorted from least to greatest. The Y-axis labels the gene mechanisms that each model is predicting the frequency of. Models are additionally color-coded by gene type, which is displayed in the legend.

Random forest models tended to explain the most variance for some of the metal resistance and biocide resistance gene types, with 7 out of the top 10 highest r-squared values being either metal or biocide gene classes. Random forest provided extremely strong models for metronidazole, chromium, and sodium resistance. Its weakest models were for mupirocin, iron– and strangely enough– biocide & metal resistance.

XGBoost Models

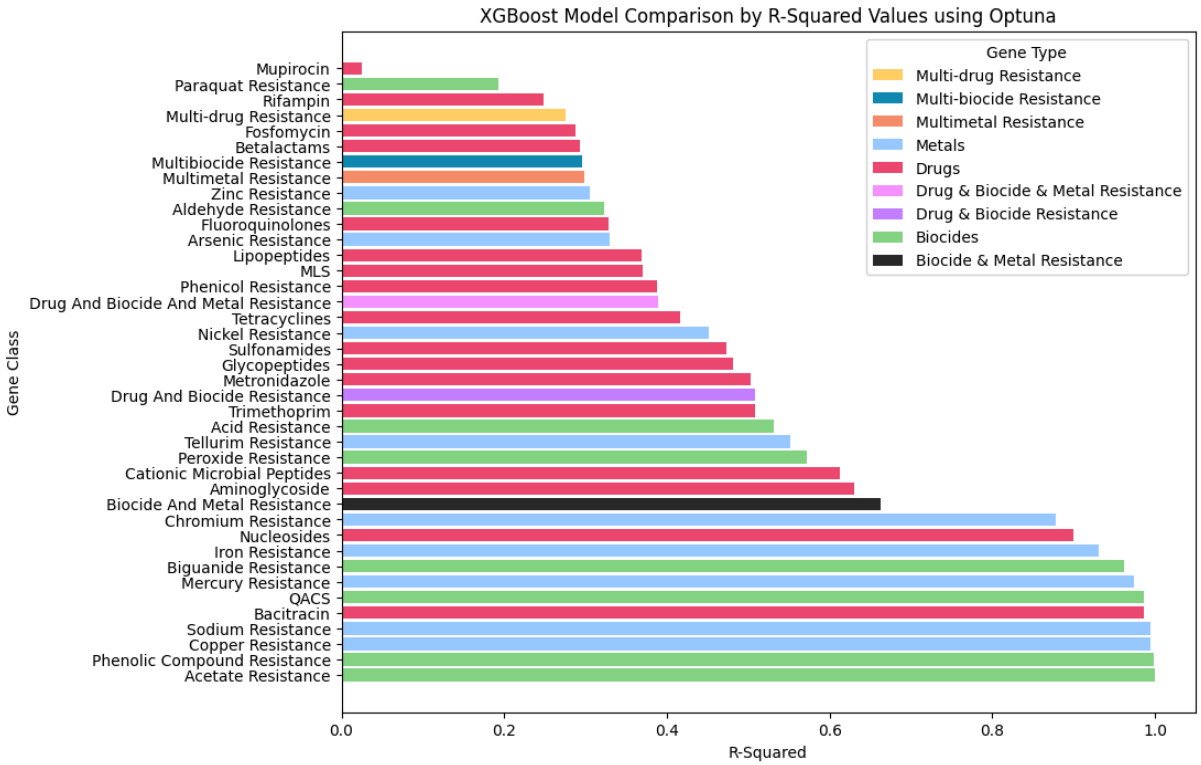

Figure 5: **R-squared Values for XGBoost Models.** Models are compared via the absolute values of their r-squared values displayed on the x-axis and sorted from least to greatest. The Y-axis labels the gene mechanisms that each model is predicting the frequency of. Models are additionally color-coded by gene type, which is displayed in the legend.

XGBoost models tended to explain the most variance for the biocide resistance gene type, with its top two highest r-squared values being biocide resistance models. XGBoost models provided extremely strong models for several AMR gene classes, including acetate resistance, phenolic compound resistance, and copper resistance. Its weakest model– by a large margin– was for mupirocin.

Lasso Models

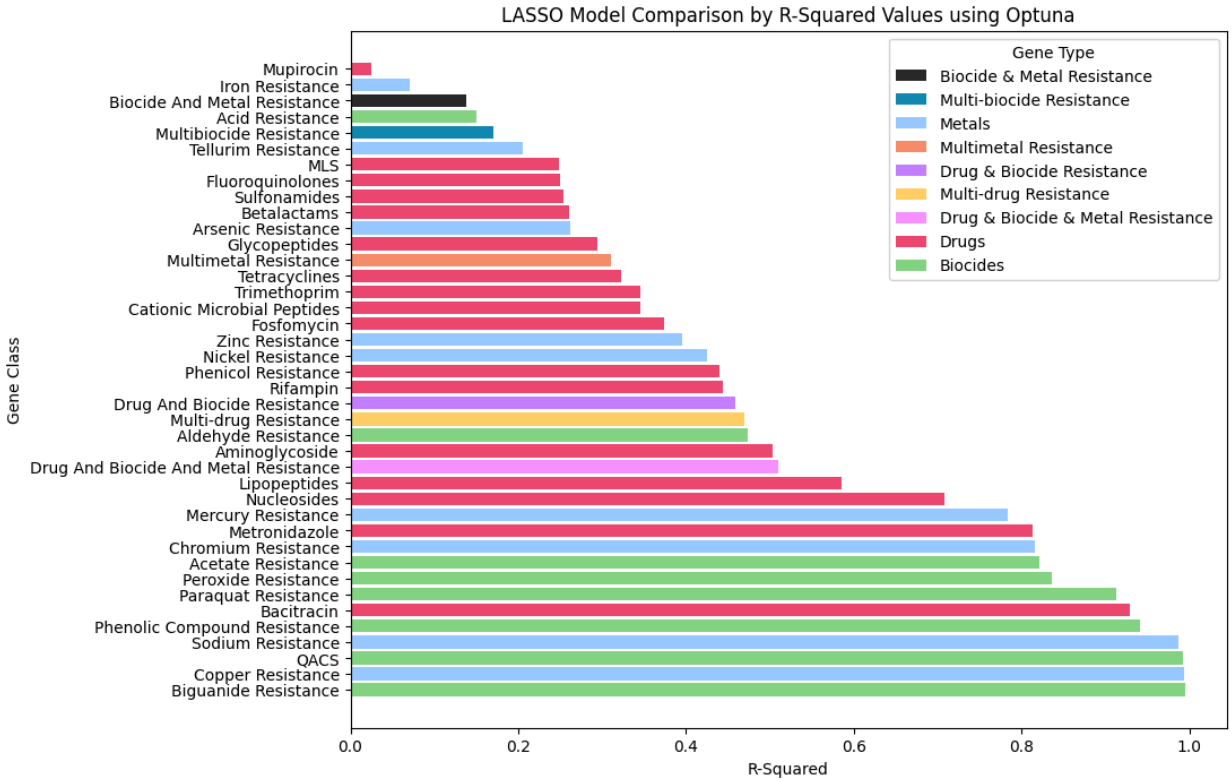

Figure 6: R-squared Values for Lasso Models. Models are compared via the absolute values of their r-squared values displayed on the x-axis and sorted from least to greatest. The Y-axis labels the gene mechanisms that each model is predicting the frequency of. Models are additionally color-coded by gene type, which is displayed in the legend.

Lasso modified linear regression models tended to explain the most variance for the biocide gene type, with 6 out of the top 10 highest r-squared values being biocide gene classes. Lasso modified linear regression models were weaker overall than XGBoost models but displayed extremely high correlation coefficient values for biguanide resistance, copper resistance, and QACs (quaternary ammonium compounds). Its weakest models were for mupirocin, iron, and biocide & metal resistance.

PCA Plot

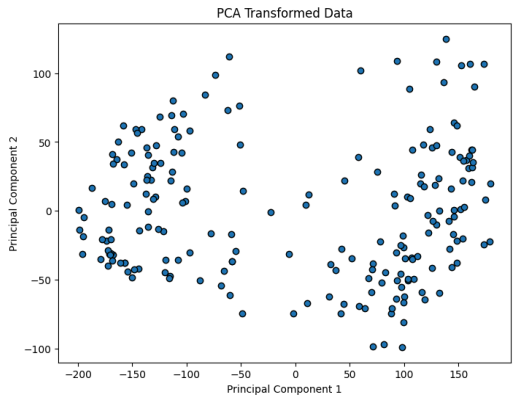

Figure 7: PCA Plot of Counts Data. The graph is incomplete as we have not yet found a way to assign sample indicators to points, rendering most of this current PCA information indecipherable.

In order to complete the PCA model created for this data, the programming language R has been considered specifically for principal component analysis. Further research must be done on implementing this study's data for PCA in R.

T-SNE Plots

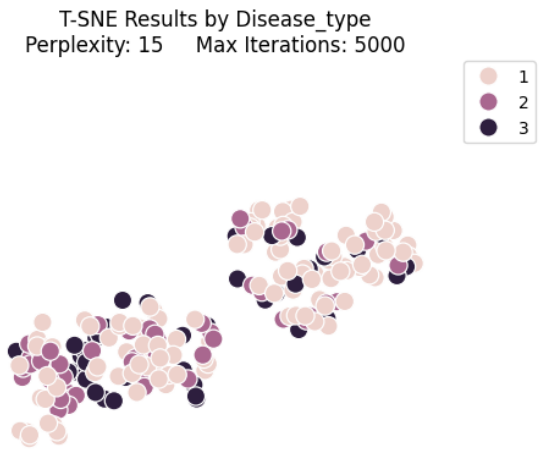

Figure 8: Example Disease Type Singular T-SNE Plot of Counts Data. The points are labelled by disease type, with the number 1 representing patient samples with rCDI, 2 being Melanoma, and 3 being MDRB (Multi-drug resistant bacteria).

T-SNE plots were generated for the counts data and then color coded to correspond to specific sample predictor columns. Plots exist for every predictor column and can be found at the Github link provided in *Software Availability*. A lack of any major trends were identified, with the t-SNE plots forming 2 roughly separate clusters that did not distinguish between any of the predictors that were provided. More refinement may be necessary to determine why the plots were clustered in the way they are.

SHAP Sample Waterfall Plots

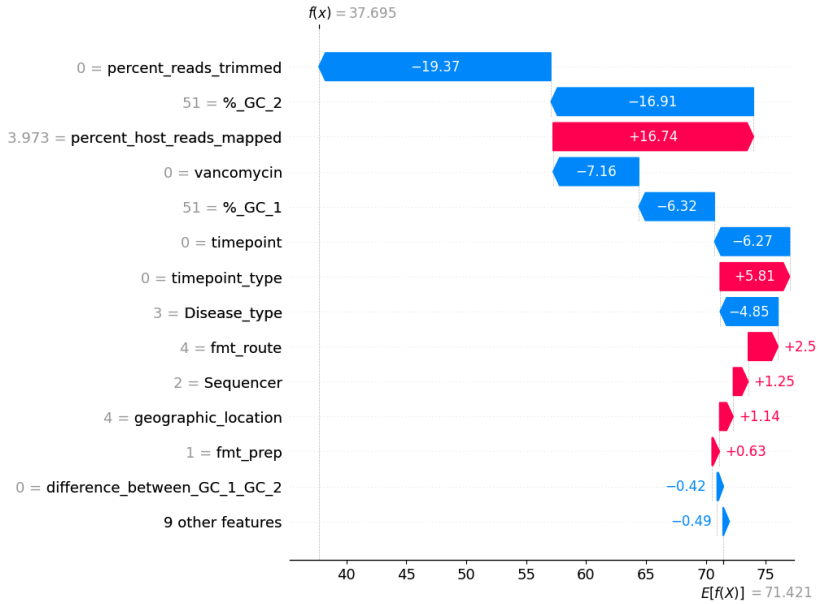

Figure 9: Example SHAP Waterfall Plot for XGBoost QACs Sample #204. This waterfall plot shows the individual impact of each feature on the AMR gene class abundance calculation (f(x)) for the sample. E[f(x)] represents the average AMR gene class abundance for all XGBoost QAC samples and how the ML model derived the AMR gene class abundance from the average.

SHAP waterfall plots can be generated for each sample using the code provided in *Software Availability* for further individual sample analysis if desired.

SHAP Beeswarm Plot

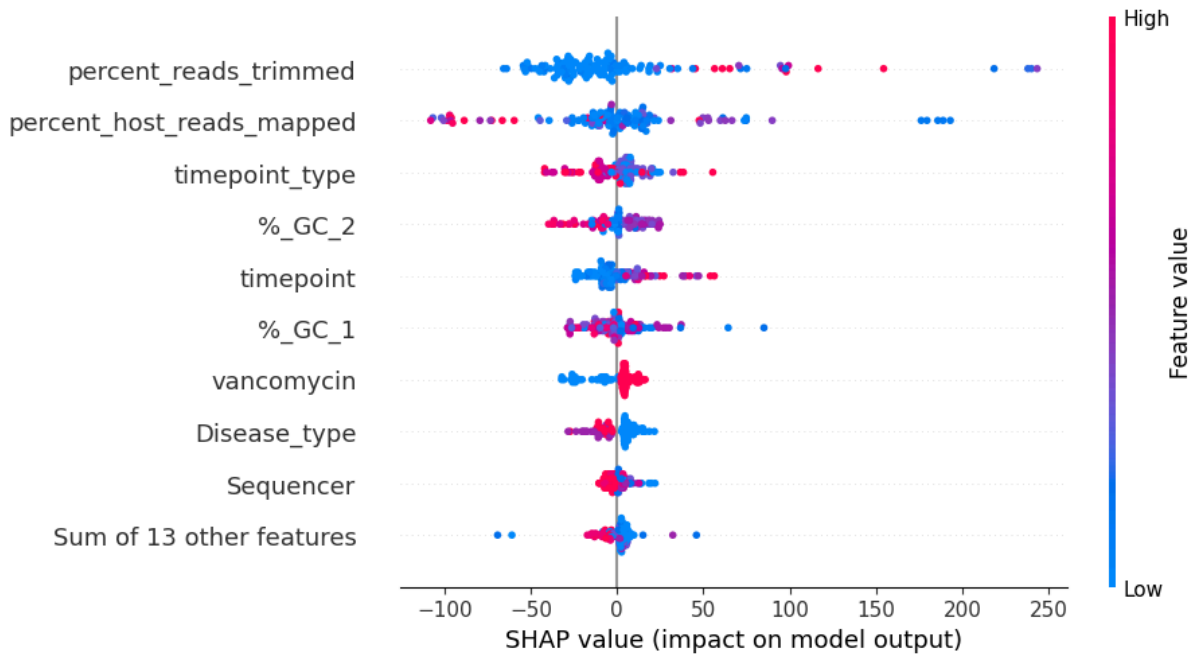

Figure 10: SHAP Beeswarm Plot for the XGBoost model of QACs. This beeswarm plot shows the impact of each feature on the model output, with the colors rating the overall value of the feature for each sample of a model and the y axis displaying the SHAP value itself while the x axis displays the list of features sorted from most to least relevant.

Using SHAP, a beeswarm plot can be generated using the code provided in *Software Availability* for each supervised machine learning method's optimal model for each AMR gene class depicting feature importance across samples.

Discussion

Interpretation of Findings

This study revealed promising data regarding the applications of Random Forest, XGBoost, and LASSO machine learning models to AMR gene class data gathered from FMT patients. Since the machine learning methods were sorted by r-squared value, we're able to compare the efficacy of different models for different gene classes. Overall, the XGBoost method generated the largest number of models with the highest correlation coefficient values, demonstrating XGBoost as a promising tool for future AMR gene research, especially for regression problems. Additionally, the LASSO method was able to compete with XGBoost as another strong method for AMR gene prediction using regression. Surprisingly enough, Random Forest, a well known and highly regarded ML method in AMR research, was found to predict quite poorly with the majority of AMR gene classes. We suspect this is the case due to the regressive nature of the prediction, though further research is needed to expand on the exact reason why.

Contributions to AMR Research

Model efficiency data can be used to inform the creation of improved models for predicting specific AMR genes. Method comparisons have pinpointed promising areas of future study in LASSO and weak points of current Random Forest methodology with regression problems. Strong models for common AMR genes associated with currently AMR drugs (e.g. bacitracin, QACs, metronidazole), when further developed, could potentially be used in predictive scenarios.

Limitations & Future Work

Due to the nature of AMR gene class grouping, models for gene classes that contain numerous subclasses and/or mechanisms may perform differently at various levels of gene classification. Refinement of models programmed in this study could potentially provide increasingly accurate tools for AMR gene prediction. Additionally, the creation of models for individual genes or gene mechanisms can amend knowledge gaps produced by models created at the class level. Classification variations could also be made to capitalize on the prevalent usage of random forest in classification problems that typically yield very good models, such as predicting presence/absence at a gene level. For AMR gene classes that were not well captured by Random Forest, XGBoost, and LASSO models such as Mupirocin, other supervised machine learning methods can be tested. For AMR gene classes with well performing methods such as metronidazole, phenolic compound resistance, and acetate resistance, these models can be further analyzed to determine the highest contributing predictors for potential clinical applications. Finally, the large number of strong models generated using the LASSO method suggests more research should be done using LASSO, as there are few existing AMR ML studies involving this type of model.

Conclusion

While antimicrobial resistance remains a major global health problem, our understanding of AMR has continued to develop and with it come ideas to limit the impact of antimicrobial resistance. This study has illustrated the potential of various machine learning methods on AMR data, and provides groundwork for further usage of ML methods in the AMR space. Refining our usage of artificial intelligence will allow us to capitalize on the information that these key computational tools can provide us, giving us better and faster ways to combat AMR.

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Appendix I

Figure 11: Machine Learning Method Evaluation Table

Method Name	**Type**	**Details**	**AMR gene or microbiome Applications**	**Count Based Data**	**Zero Inflated Data**	**Observation imbalance/nested data**	**Implementation**	**Reasoning**
Random Forest	Supervised	Aggregates bootstrapped decision trees from randomly sampled subsets of data to predict outcomes. https://link.springer.com/article/10.1023/a:1010933404324	https://link.springer.com/article/10.1186/s13567-023-01141-5 Used to predict AMR susceptibility	https://onlinelibrary.wiley.com/doi/full/10.1155/2022/2833537 Is a good alternative to common statistical methods for counts data	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0216511#sec005 Canonical ML methods don't do a great job with zero inflated data unless paired with other models	Data balance is likely an issue for random forest regression https://www.cambridge.org/core/journals/environmental-data-science/article/variable-ranking-and-selection-with-random-forest-for-unbalanced-data/D00D9D74FA395B4FAC8886A84CC2FCCA and strategies are far less researched than for classification. https://pr	Scikit-Learn	Common Usage, Performs Well/State of the art [1] https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1004977 Can be used in both regression and classification problems https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0216511#sec005

						oceedings.mlr.press/v74/branco17a		
Gradient Boosting Decision Tree Regression	Supervised	Trains and revises decision trees to predict residuals based off of a gradient equation.	https://www.nature.com/articles/s43856-024-00606-8 Predicting trends in future hospital AMR https://journals.asm.org/doi/10.1128/jcm.01260-18 Predicting antimicrobial genomic features in bacteria	https://link.springer.com/article/10.1007/s10994-015-5506-z Gradient boosting models provide a better model of count data	Poisson Gradient models perform better than standard ones https://arxiv.org/pdf/2307.07771	https://ieeexplore.ieee.org/abstract/document/10807295 Very little research on XGboost regression but most classification doesn't do well with imbalanced data. One method for increasing XG boost performance in unbalanced data: https://arxiv.org/pdf/2206.09991	XGBoost	Common Usage [4], Can be used in both regression and classification problems.
Lasso (Least Absolute Shrinkage and Selection Operator)	Supervised	Regularizes a linear regression model to reduce overfitting errors. Common	Used in a similar application (predicting CRC from metagenomic	https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2016.00032	https://www.frontiersin.org/journals/genetics/articles/10.3389/fgene.2016.00032	Performs better with observation imbalance data https://journals.sa	Scikit-Learn	Common usage

		ly used in tandem with other machine learning methods [4] In contrast to ridge regression, lasso can eliminate unnecessary predictors entirely, as opposed to shrinking them. [8] https://academic.oup.com/jrsssb/article/58/1/267/7027929?login=false	data), makes model interpretation and marker extraction easy [7], also used here https://europepmc.org/article/med/33844808 to determine AMR from predictors which is exactly what im doing	/full Needs to be modified to tackle zero inflated counts based data	/full Needs to be modified to tackle zero inflated counts based data	gepub.com/doi/abs/10.1177/096228021664 3116		
Ridge Regression	Supervised	Adds a penalty determined through cross validation to a linear regression model to reduce overfitting errors by decreasing variance.		Struggling to find similar AMR studies that utilize ridge regression (most use it for binary classification). Many studies just use LASSO.				
Logistic Regression	Supervised	Classifies output by predicting the categorically dependent variable.		Not considered for this study since it is primarily a classification technique.				
Support Vector	Supervised	Determine the optimal		Not considered for this study because it is relegated to benchmarking, no longer commonly used [1], used				

Machines		hyperplane/decision boundary separating data points into different classes.		mainly for classification though has another form called SVR that can be used for regression [6]		
PCA	Unsupervised	Reduces the size of a dataset while maintaining relevant features by clustering correlated variables together.	Used here https://www.mdpi.com/2079-6382/11/2/229 to predict AMR from infection control metadata in hospitals. Used here to group children's microbiome data to predict BMI https://www.sciencedirect.com/science/article/pii/S0002916522009741	Although not a regression method for prediction, PCA provides a way to visually see how features are grouped between samples, as well as acts as a preprocessing step for regressive data.	Scikit learn	A good visualization of correlation between samples. [10] Performs well regressively here to predict BMI from gut microbiota [11], also used here to cluster AMR genes [12]
PCR	Unsupervised	Applies a linear regressor to data that has been transformed by PCA	Used here https://pmc.ncbi.nlm.nih.gov/articles/PMC6138053/	PCR is an unsupervised technique based off of PCA and is relatively unused in the realm of AMRand microbiome.	Scikit learn	See PCA

t-SNE (t-distributed Stochastic Neighbor Embedding)	Unsupervised	Clusters data based on non linear pairwise similarities.	https://distill.pub/2016/misread-tsne/ used this to interpret t-sne	https://lvdmaaten.github.io/tsne/ user guide by the creator	Used here https://arxiv.org/pdf/2504.17717	Captures non-linear relationships well, Finicky tuning [1]

Appendix II

Figure 12: Data Conversion Reference Table

Predictor	**Before**	**After**
Disease_Type	rCDI	1
	Melanoma	2
	MDRB	3
Donor_pre_post	Pre-FMT or Pre-ABX/FMT	0
	Post-FMT	1
Geographic_Location	USA	1
	Canada	2
	Italy	3
	Switzerland	4
	Israell	5
FMT_Route	Enteroscopy & Colonoscopy	0
	Colonoscopy	1
	Colonoscopy or Nasogastric Tube	2
	Colonoscopy & 12 capsules	3
	Nasogastric Tube	4
	Nasoduodendal Tube	5
	Oral Capsules	6
	30 Capsules	7

FMT_Prep	Fresh	0
	Frozen	1
Bowel_Preparation	Unspecified	0
	Macrogol/Macrogel	1
	None	2
Vancomycin/Colistin/Neomystin/Other_Abx	FALSE	0
	TRUE	1
median_sequence_quality_above_30_1/median_sequence_quality_above_30_2	N	0
	Y	1